

第一題

$$1.19x_1 + 2.11x_2 - 100x_3 + x_4 = 1.12$$

$$14.2x_1 - 0.112x_2 + 12.2x_3 - 3.44x_4 = -1$$

$$100x_2 - 99.9x_3 + 2.15x_4 = 0$$

$$15.3x_1 + 0.110x_2 - 13.1x_3 - 4.16x_4 = 1$$

Solution:

$$x_1 = 0.176825$$

$$x_2 = 0.012692$$

$$x_3 = 0.020654$$

$$x_4 = -0.004114$$

第二題

Find the inverse of matrix A:

A = [

[4, 1, -1, 0],

[1, 3, -1, 0],

[-1, -1, 6, 2],

[0, 0, 2, 5]

]

Inverse of matrix A:

[[0.27027027 -0.08108108 0.02702703 -0.01081081]

[-0.08108108 0.35135135 0.05405405 -0.02162162]

[0.02702703 0.05405405 0.18918919 -0.07567568]

[-0.01081081 -0.02162162 -0.07567568 0.23027027]]

第三題

Solve the tridiagonal system using Crout factorization:

$$\begin{bmatrix} 3 & -1 & 0 & 0 \\ -1 & 3 & -1 & 0 \\ 0 & -1 & 3 & -1 \\ 0 & 0 & -1 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ 4 \\ 1 \end{bmatrix}$$

Solution:

$$x_1 = 1.000000$$

$$x_2 = 1.000000$$

$$x_3 = 1.000000$$

$$x_4 = 0.666667$$